

```
!pip install boto3
!pip install ruff
!pip install bandit
!pip install mypy
```

```
Requirement already satisfied: boto3 in /usr/local/lib/python3.12/dist-packages (1.40.41)
Requirement already satisfied: botocore<1.41.0,>=1.40.41 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.40.41)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.0.1)
Requirement already satisfied: s3transfer<0.15.0,>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from boto3) (0.14.0)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/local/lib/python3.12/dist-packages (from botocore<1.41.0,>=1.40.41) (2.9.0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /usr/local/lib/python3.12/dist-packages (from botocore<1.41.0,>=1.40.41) (2.2.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.41.0,>=1.40.41) (1.17.0)
Requirement already satisfied: ruff in /usr/local/lib/python3.12/dist-packages (0.13.0)
Collecting bandit
  Downloading bandit-1.8.6-py3-none-any.whl.metadata (6.9 kB)
Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from bandit) (6.0.2)
Collecting stevedore>=1.20.0 (from bandit)
  Downloading stevedore-5.5.0-py3-none-any.whl.metadata (2.2 kB)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from bandit) (13.9.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->bandit) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->bandit) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->bandit) (0.1.2)
Downloading bandit-1.8.6-py3-none-any.whl (133 kB)
----- 133.8/133.8 kB 3.8 MB/s eta 0:00:00
Downloading stevedore-5.5.0-py3-none-any.whl (49 kB)
----- 49.5/49.5 kB 3.5 MB/s eta 0:00:00
Installing collected packages: stevedore, bandit
Successfully installed bandit-1.8.6 stevedore-5.5.0
Collecting mypy
  Downloading mypy-1.18.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.2 kB)
Requirement already satisfied: typing_extensions>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from mypy) (4.15.0)
Collecting mypy_extensions>=1.0.0 (from mypy)
  Downloading mypy_extensions-1.1.0-py3-none-any.whl.metadata (1.1 kB)
Collecting pathspec>=0.9.0 (from mypy)
  Downloading pathspec-0.12.1-py3-none-any.whl.metadata (21 kB)
Downloading mypy-1.18.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (13.3 MB)
----- 13.3/13.3 MB 116.2 MB/s eta 0:00:00
Downloading mypy_extensions-1.1.0-py3-none-any.whl (5.0 kB)
Downloading pathspec-0.12.1-py3-none-any.whl (31 kB)
Installing collected packages: pathspec, mypy_extensions, mypy
Successfully installed mypy-1.18.2 mypy_extensions-1.1.0 pathspec-0.12.1
```

```
import boto3
import json
import tempfile
import subprocess

# ---- AWS Bedrock direct credentials ----
AWS_ACCESS_KEY = "AKIAVSX4Z23PH4R4CRWQ"
AWS_SECRET_KEY = "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"
AWS_REGION = "us-east-1" # region where Claude 3 Sonnet is available
MODEL_ID = "anthropic.claude-3-sonnet-20240229-v1:0"

# ---- Prompt template ----
FEW_SHOT_TEMPLATE = """
You are a professional Python developer. Implement functions according to the spec. Examples show desired style.

Example 1:
Spec: Create function add(a:int,b:int)->int that returns a+b.
Code:
def add(a: int, b: int) -> int:
    \"\"\"Return sum of a and b.\"\"\"
    return a + b

Now implement:
Spec:
{spec_text}

Code:
\"\"\".strip()

# ---- Invoke Bedrock model ----
def invoke_model(prompt: str) -> str:
    rt = boto3.client(
        "bedrock-runtime",
        region_name=AWS_REGION,
        aws_access_key_id=AWS_ACCESS_KEY,
        aws_secret_access_key=AWS_SECRET_KEY
    )
    body = {
        "anthropic_version": "bedrock-2023-05-31",
        "max_tokens": 800,
        "temperature": 0.2,
```

```

        "messages": [{"role": "user", "content": prompt}],
    }
    resp = rt.invoke_model(
        modelId=MODEL_ID,
        body=json.dumps(body),
        contentType="application/json"
    )
    obj = json.loads(resp["body"].read())
    if "content" in obj and len(obj["content"]) > 0:
        return obj["content"][0].get("text", "")
    return ""

# ---- Generate code ----
def generate_code(spec_text: str):
    prompt = FEW_SHOT_TEMPLATE.format(spec_text=spec_text)
    code_text = invoke_model(prompt)
    # remove code fences
    if code_text.startswith("```"):
        code_text = code_text.strip("```").strip()
    return code_text

# ---- Optionally run static checks ----
def run_linters(code_path: str):
    results = {}
    results["ruff"] = subprocess.run(["ruff", code_path], capture_output=True, text=True)
    results["bandit"] = subprocess.run(["bandit", "-r", code_path], capture_output=True, text=True)
    results["mypy"] = subprocess.run(["mypy", code_path], capture_output=True, text=True)
    return results

# ---- Main ----
if __name__ == "__main__":
    spec = "Write a Python function def factorial(n:int)->int that returns factorial of n (n>=0). Raise ValueError if n<0."
    code = generate_code(spec)

    # Save code to a temporary file
    tmp_file = tempfile.NamedTemporaryFile(delete=False, suffix=".py")
    tmp_file.write(code.encode("utf-8"))
    tmp_file.close()

    print("=== Generated Code ===")
    print(code)

    # Run lint/security checks
    lint_results = run_linters(tmp_file.name)
    print("\n=== Lint / Security Checks ===")
    for tool, result in lint_results.items():
        print(f"--- {tool} ---")
        print(result.stdout)
        print(result.stderr)

```python
print(factorial(0)) # Output: 1
print(factorial(5)) # Output: 120
print(factorial(10)) # Output: 3628800
try:

```

```
Low: 0
Medium: 0
High: 0
Total issues (by confidence):
Undefined: 0
Low: 0
Medium: 0
High: 0
Files skipped (1):
/tmp/tmp5m_hmjrm.py (syntax error while parsing AST from file)

[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.12.11

--- mpy ---
/tmp/tmp5m_hmjrm.py:1: error: Unterminated string literal (detected at line 1) [syntax]
Found 1 error in 1 file (errors prevented further checking)
```